

Introduction to Agentic AI

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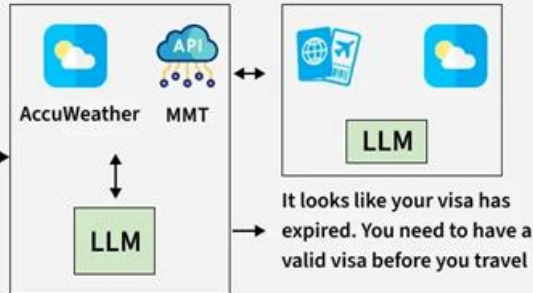
AI Use cases: Gen AI vs AI Agent vs Agentic AI

Generative AI

What is the price of Emirates flight from New York to Delhi tomorrow? → **LLM** → I do not know the latest price. My knowledge is cut off till Dec 2024

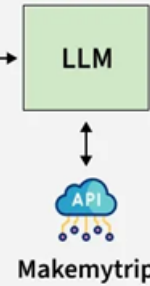
Agentic AI

Book me a flight for my 7-day trip to New Delhi in May. The weather should be sunny all days, budget below \$1600, and no layovers.



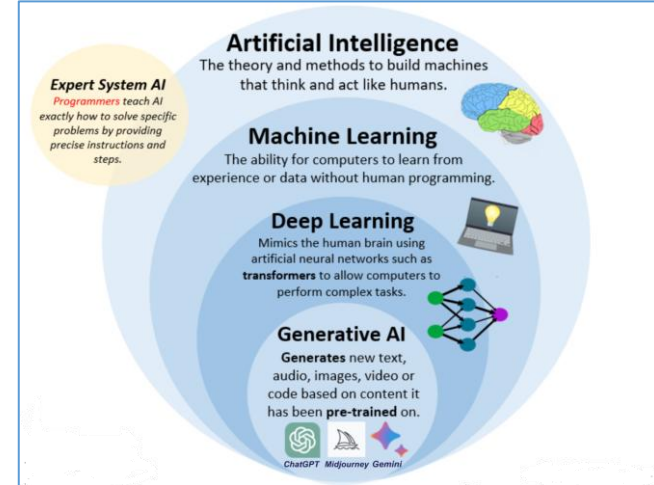
AI Agent

Tell me the cheapest Emirates flight from New York to Delhi tomorrow → **LLM** → The price is 1200 \$ - economy and 3400 \$ business class



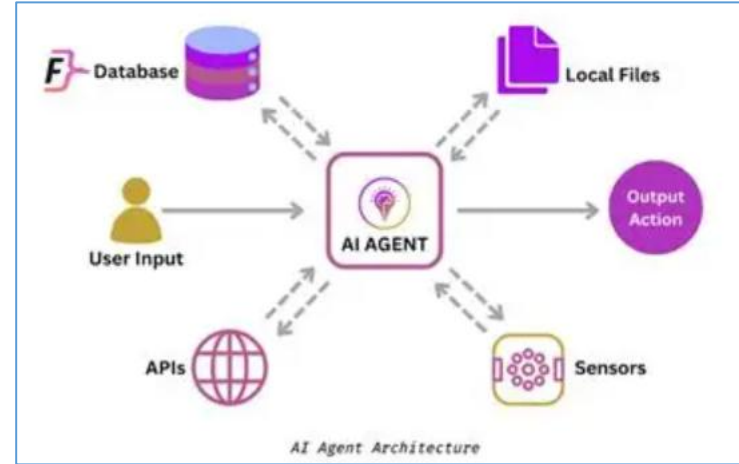
What is Gen AI?

- Generative AI is a branch of artificial intelligence that focuses on creating new content such as text, images, code, music and video etc.
- Gen AI mainly uses models that are based upon transformers or GANs architecture.
- These models generate original outputs that are often indistinguishable from human-created content.
- These models use techniques like deep learning and neural networks to generate output.
- Gen AI is used in tools like ChatGPT, Claude, Gemini and DALL-E & is applied in automation, chatbots and personalization.



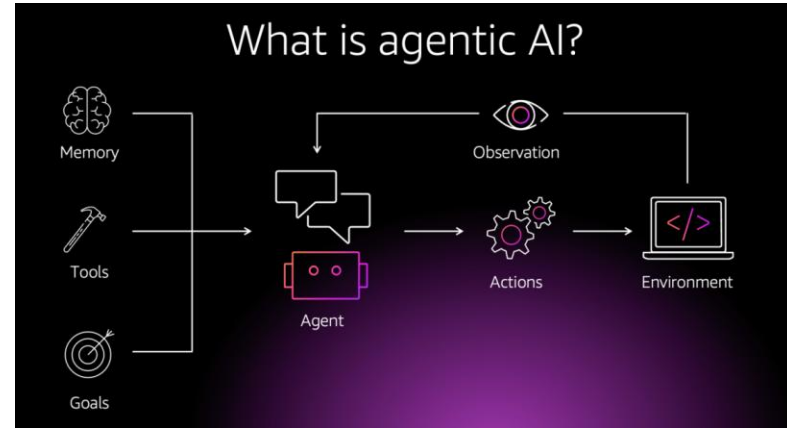
What is an AI Agent?

- AI agents are advanced, autonomous, or semi-autonomous software systems that analyze, plan, and execute tasks independently.
- When used with proper governance, they leverage artificial intelligence to process information, make decisions, and perform actions while adhering to established business rules.
- Most importantly, these intelligent agents can optimize operations through continuous learning and adaptation, improving efficiency and quality of output over time.



What is an Agentic AI?

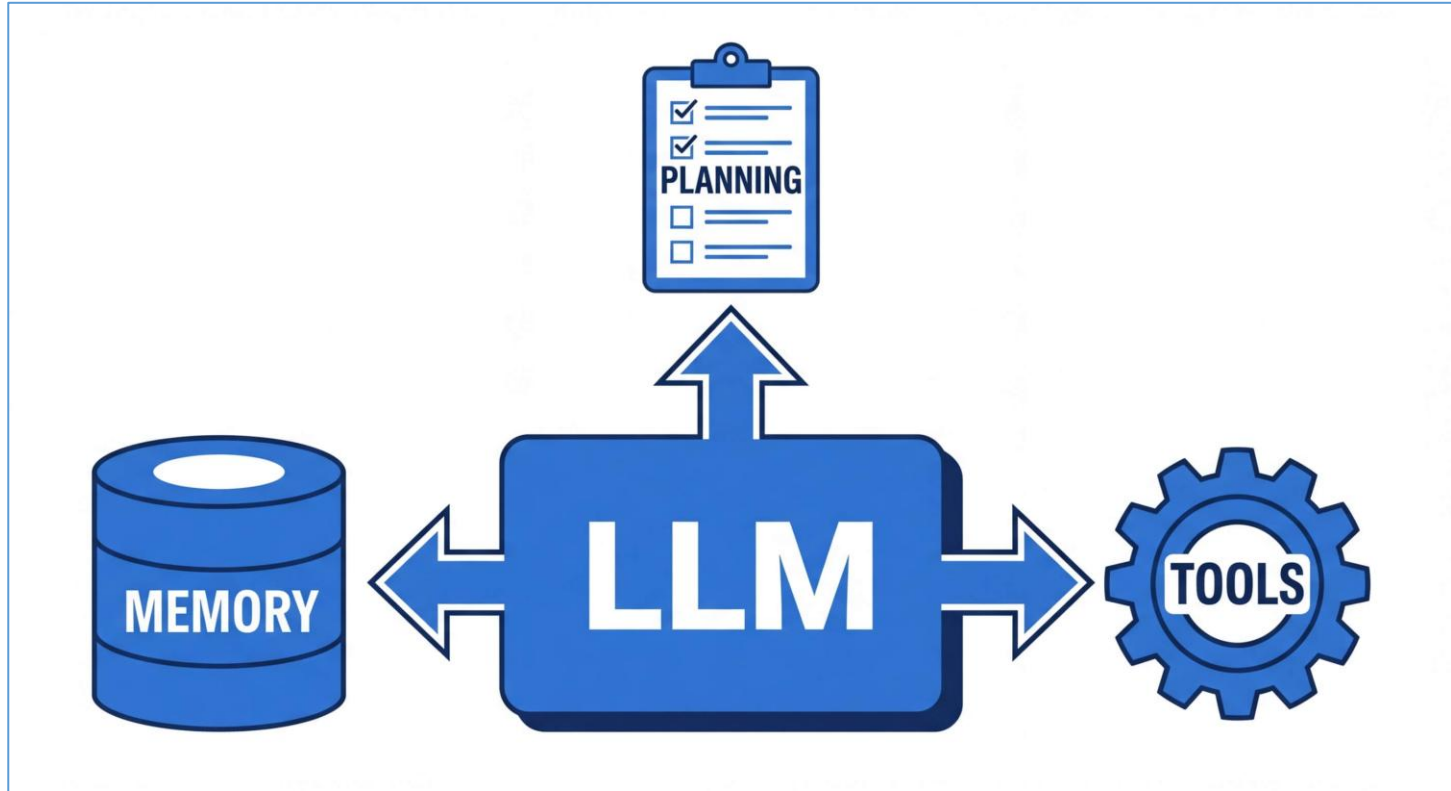
- Agentic AI refers to autonomous artificial intelligence systems designed to achieve specific goals with minimal human supervision.
- Unlike traditional AI, which operates within predefined constraints and requires human intervention, agentic AI exhibits autonomy, adaptability, and goal-driven behavior.
- It uses AI agents—machine learning models that mimic human decision-making—to solve problems in real-time by interacting with their environment.



Gen AI vs AI Agent vs Agentic AI

Feature	Gen AI	AI Agent	Agentic AI
Primary Function	Generates new content (text, images, videos, code)	Performs specific predefined tasks automatically	Plans, reasons and acts independently toward a goal
Dependency	Works based on trained data	Depends on predefined rules and APIs	Uses reasoning, planning and feedback loops
Tool Access	No real-time access to tools or APIs	Limited access to tools for specific actions	Dynamic access to multiple tools and APIs
Learning Capability	Learns from historical data (not real-time)	No continuous learning, follows instructions	Continuously adapts to context and outcomes
Example Use Case	Writing articles, creating designs	Fetching flight data or sending alerts	Booking flights based on preferences and constraints
Autonomy Level	Low	Medium	High
Example Tools/Models	ChatGPT, DALL·E, Midjourney	Customer service bots, automation systems	AutoGPT, Devin AI, Gemini 2.0 with planning capabilities

Core Components of an Agentic AI System



Core Components of an Agentic AI System

The Reasoning Engine (LLM)

- At the core of every agent is a foundational Large Language Model(LLM).
- The engine is responsible for parsing user instructions, synthesizing information, and determining the sequence of operations required.
- For agentic workflows, the model must possess strong instruction-following capabilities and zero-shot reasoning proficiencies.

Core Components of an Agentic AI System

Memory Subsystems:

Agents require memory to maintain context over prolonged execution cycles. This is typically bifurcated into:

- *Short-Term Memory*: Implemented via the immediate context window of the LLM. It stores the recent conversation history, the current plan, and the immediate observations from executed tools.
- *Long-Term Memory*: Implemented via external data stores, predominantly Vector Databases (e.g., Pinecone, Milvus, or local implementations like ChromaDB). Information is converted into dense vector embeddings and retrieved using mathematical similarity metrics.

Core Components of an Agentic AI System

Tools or Actuators

- Without tools, an agent is trapped within its context window. Tools (or actuators) are the deterministic Python functions, REST APIs, GraphQL endpoints, or shell environments that the agent can invoke.
- When an agent decides to use a tool, it formats the required parameters (often as a JSON object), passes them to the tool's execution environment, and ingests the resulting output back into its memory as an observation.

Popular Frameworks and Libraries for Agentic AI

Framework	Primary Focus	Multi-Agent Support	Ideal Use Case
LangChain / LangGraph	General purpose LLM orchestration and data ingestion.	Moderate (via LangGraph)	Building complex, stateful single-agent loops and Retrieval-Augmented Generation (RAG) pipelines.
LlamaIndex	Data connection, indexing, and advanced retrieval strategies.	Basic	Data-heavy agents that need to query massive enterprise document repositories accurately.
CrewAI	Role-playing autonomous multi-agent orchestration.	High	Simulating organizational workflows where distinct agents (e.g., Researcher, Writer, QA) collaborate.
Microsoft AutoGen	Conversational multi-agent systems with robust code execution sandboxing.	High	Engineering and coding agents that write, execute, and debug code autonomously.

Thank you!!